Math 21 — Algebra and its Functions

Middlesex School Mathematics | Textbook: OpenStax Intermediate Algebra 2e (Marecek and Mathis, 2020)

COURSE AT A GLANCE

Review Content

REVIEW

3.1 Graph Linear Equations in Two Variables
3.2 Slope of a Line
3.3 Find the Equation of a Line
5.2 Properties of Exponents and Scientific Notation

UNIT LEARNING OBJECTIVES

I can determine and interpret the slope of a line.
I can graph a line from an equation and write an equation from given information.
I can apply a linear function to a context.
I can identify the solution of a linear system as an intersection point.
I can solve a simple two-variable linear system using an appropriate method.
I can simplify an expression using basic positive-integer exponent rules.

Functions and Function Interpretation

REQUIRED

3.5 Relations and Functions
3.6 Graphs of Functions

UNIT LEARNING OBJECTIVES

I can evaluate a function from a formula, graph, or table.
I can determine whether a relation is a function.
I can determine domain and range from a graph or table.
I can solve for an input that produces a specified output.
I can interpret inputs, outputs, domain, and range in context.
I can calculate and interpret average rate of change.
I can compare functions represented by formulas, graphs, or tables.

Polynomials and Factoring

REQUIRED

5.1 Add and Subtract Polynomials
5.3 Multiply Polynomials
5.4 Dividing Polynomials
6.1 Greatest Common Factor and Factor by Grouping
6.2 Factor Trinomials
6.3 Factor Special Products

UNIT LEARNING OBJECTIVES

I can add and subtract polynomials.
I can multiply polynomials.
I can factor out a greatest common factor.
I can factor a quadratic trinomial.
I can factor a difference of squares and other required special cases.
I can use factoring to determine the zeros of a polynomial.

Rational Expressions and Equations

REQUIRED

7.1 Simplify Rational Expressions
7.2 Multiply and Divide Rational Expressions
7.3 Add and Subtract Rational Expressions with a Common Denominator
7.4 Add and Subtract Rational Expressions with Unlike Denominators
7.5 Solve Rational Equations

UNIT LEARNING OBJECTIVES

I can state excluded values for a rational expression.

I can simplify a rational expression while preserving its restrictions.

I can multiply and divide rational expressions.

I can add and subtract rational expressions using a common denominator.

I can solve a rational equation and reject extraneous solutions.

Exponents and Radicals

REQUIRED

8.1 Simplify Expressions with Roots
8.2 Simplify Radical Expressions
8.3 Simplify Rational Exponents
8.4 Add, Subtract, and Multiply Radical Expressions

UNIT LEARNING OBJECTIVES

I can simplify expressions containing negative exponents.

I can rewrite expressions between radical and rational-exponent forms.

I can simplify radical expressions.

I can apply exponent rules to expressions with integer and rational exponents.

I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.

Quadratic Functions and Equations

REQUIRED

9.1 Solve Quadratic Equations Using the Square Root Property
9.2 Solve Quadratic Equations by Completing the Square
9.3 Solve Quadratic Equations Using the Quadratic Formula
9.4 Solve Equations in Quadratic Form
9.5 Solve Applications of Quadratic Equations
9.6 Graph Quadratic Functions Using Properties
9.7 Graph Quadratic Functions Using Transformations

UNIT LEARNING OBJECTIVES

I can identify key features of a quadratic function from its graph or equation.

I can connect standard, factored, and vertex forms to features of a parabola.

I can solve a quadratic equation by factoring.

I can solve a quadratic equation using the square-root property.

I can solve a quadratic equation by completing the square.

I can solve a quadratic equation using the quadratic formula.

I can choose an efficient method for solving a quadratic equation.

I can graph a quadratic function using its vertex, intercepts, and symmetry.

I can create and interpret a quadratic model for a projectile, area, or optimization problem.

Extension Content

EXTENSION

9.3 Solve Quadratic Equations Using the Quadratic Formula

UNIT LEARNING OBJECTIVES

I can perform basic arithmetic with complex numbers.

I can express the nonreal solutions of a quadratic equation using complex numbers.

IN-DEPTH CURRICULUM

SKILL PROGRESSION I Introduce R Reinforce D Deepen A Apply

Review Content REVIEW

Textbook coverage: 3.1 Graph Linear Equations in Two Variables; 3.2 Slope of a Line; 3.3 Find the Equation of a Line; 5.2 Properties of Exponents and Scientific Notation

M21-REV-001

I can determine and interpret the slope of a line.

Textbook: 3.2 Slope of a Line

SUPPORTING SKILLS

- R Determine slope from a graph.
  SK-LIN-SLOPE-GRAPH · first introduced in Math 12
- R Calculate slope from two points.
  SK-LIN-SLOPE-POINTS · first introduced in Math 12
- R Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M21-REV-002

I can graph a line from an equation and write an equation from given information.

Textbook: 3.1 Graph Linear Equations in Two Variables; 3.3 Find the Equation of a Line

SUPPORTING SKILLS

- R Convert among common forms of a linear equation.
  SK-LIN-FORM-CONVERT · first introduced in Math 12
- R Use slope to generate additional points on a line.
  SK-LIN-GRAPH-SLOPE · first introduced in Math 12
- R Plot an intercept accurately on a coordinate plane.
  SK-LIN-PLOT-INTERCEPT · first introduced in Math 12
- A Calculate slope from two points.
  SK-LIN-SLOPE-POINTS · first introduced in Math 12

M21-REV-003

I can apply a linear function to a context.

Textbook: 3.3 Find the Equation of a Line

SUPPORTING SKILLS

- I Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET
- R Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M21-REV-004

I can identify the solution of a linear system as an intersection point.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- R Locate and interpret the intersection of two lines.
  SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12

M21-REV-005

I can solve a simple two-variable linear system using an appropriate method.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- R Check an ordered pair in both equations of a system.
  SK-SYS-CHECK · first introduced in Math 12
- R Add equations to eliminate a variable.
  SK-SYS-ELIMINATE · first introduced in Math 12
- R Locate and interpret the intersection of two lines.
  SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12
- R Substitute an equivalent expression for a variable.
  SK-SYS-SUBSTITUTE · first introduced in Math 12

M21-REV-006

I can simplify an expression using basic positive-integer exponent rules.

Textbook: 5.2 Properties of Exponents and Scientific Notation

SUPPORTING SKILLS

- R Apply the power rule for exponents.
  SK-EXP-POWER · first introduced in Math 12
- R Apply the product rule for exponents.
  SK-EXP-PRODUCT · first introduced in Math 12
- R Apply the quotient rule for exponents.
  SK-EXP-QUOTIENT · first introduced in Math 12

Functions and Function Interpretation REQUIRED

Textbook coverage: 3.5 Relations and Functions; 3.6 Graphs of Functions

M21-FUN-001

I can evaluate a function from a formula, graph, or table.

Textbook: 3.5 Relations and Functions; 3.6 Graphs of Functions

SUPPORTING SKILLS

- A Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- D Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A Identify the input and output represented by function notation.
  SK-FUNC-NOTATION-READ · first introduced in Math 12

M21-FUN-002

I can determine whether a relation is a function.

Textbook: 3.5 Relations and Functions; 3.6 Graphs of Functions

SUPPORTING SKILLS

- D Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D Apply the vertical line test to a graph.
  SK-FUNC-VERTICAL-LINE · first introduced in Math 12

M21-FUN-003

I can determine domain and range from a graph or table.

Textbook: 3.5 Relations and Functions; 3.6 Graphs of Functions

SUPPORTING SKILLS

- D Read domain and range from a graph.
  SK-FUNC-DOMAIN-RANGE-GRAPH · first introduced in Math 12
- D Read domain and range from a table.
  SK-FUNC-DOMAIN-RANGE-TABLE · first introduced in Math 12

M21-FUN-004

I can solve for an input that produces a specified output.

Textbook: 3.5 Relations and Functions; 3.6 Graphs of Functions

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I Solve for an input associated with a specified output.
  SK-FUNC-SOLVE-INPUT

M21-FUN-005

I can interpret inputs, outputs, domain, and range in context.

Textbook: 3.5 Relations and Functions; 3.6 Graphs of Functions

SUPPORTING SKILLS

- D Restrict domain to values meaningful in a context.
  SK-FUNC-CONTEXT-DOMAIN · first introduced in Math 12
- D Interpret the range of a function in a context.
  SK-FUNC-CONTEXT-RANGE · first introduced in Math 12
- D Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12

M21-FUN-006

I can calculate and interpret average rate of change.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- I Calculate average rate of change over an interval.
  SK-FUNC-AROC
- I Interpret average rate of change with contextual units.
  SK-FUNC-AROC-INTERPRET
- D Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

M21-FUN-007

I can compare functions represented by formulas, graphs, or tables.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- I Compare functions shown in different representations.
  SK-FUNC-COMPARE-REP
- I Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH
- I Use tabular patterns to predict graph shape and behavior.
  SK-REP-TABLE-GRAPH

Polynomials and Factoring REQUIRED

Textbook coverage: 5.1 Add and Subtract Polynomials; 5.3 Multiply Polynomials; 5.4 Dividing Polynomials; 6.1 Greatest Common Factor and Factor by Grouping; 6.2 Factor Trinomials; 6.3 Factor Special Products

---

M21-POL-001

I can add and subtract polynomials.

Textbook: 5.1 Add and Subtract Polynomials; 5.4 Dividing Polynomials

SUPPORTING SKILLS

- A Identify and combine like terms.
  SK-ALG-LIKE-TERMS · inherited from middle school
- D Identify like polynomial terms by variable part and exponent.
  SK-POLY-LIKE-TERMS · first introduced in Math 12
- R Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-002

I can multiply polynomials.

Textbook: 5.3 Multiply Polynomials; 5.4 Dividing Polynomials

SUPPORTING SKILLS

- A Apply the distributive property in either direction.
  SK-ALG-DISTRIBUTE · inherited from middle school
- D Multiply two binomials using distribution.
  SK-POLY-BINOMIAL-MULTIPLY · first introduced in Math 12
- D Multiply a monomial by a polynomial.
  SK-POLY-DISTRIBUTE · first introduced in Math 12
- A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-003

I can factor out a greatest common factor.

Textbook: 6.1 Greatest Common Factor and Factor by Grouping

SUPPORTING SKILLS

- R Verify a factorization by multiplication.
  SK-FACTOR-CHECK · first introduced in Math 12
- D Factor the greatest common monomial factor from a polynomial.
  SK-FACTOR-GCF · first introduced in Math 12
- A Determine the numerical and variable greatest common factor of polynomial terms.
  SK-POLY-GCF · first introduced in Math 12
- A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-004

I can factor a quadratic trinomial.

Textbook: 6.1 Greatest Common Factor and Factor by Grouping; 6.2 Factor Trinomials

SUPPORTING SKILLS

- A Verify a factorization by multiplication.
  SK-FACTOR-CHECK · first introduced in Math 12
- R Factor a quadratic trinomial with a nonunit leading coefficient.
  SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- R Factor a monic quadratic trinomial.
  SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-005

I can factor a difference of squares and other required special cases.

Textbook: 6.3 Factor Special Products

SUPPORTING SKILLS

- A Verify a factorization by multiplication.
  SK-FACTOR-CHECK · first introduced in Math 12
- I Factor a difference of two squares.
  SK-FACTOR-DIFF-SQUARES
- A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-006

I can use factoring to determine the zeros of a polynomial.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- A Factor a difference of two squares.
  SK-FACTOR-DIFF-SQUARES · first introduced in Math 21
- A Factor a quadratic trinomial with a nonunit leading coefficient.
  SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- A Factor a monic quadratic trinomial.
  SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- I Apply the zero-product property.
  SK-FACTOR-ZERO-PRODUCT
- A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
  SK-POLY-VOCABULARY · first introduced in Math 12

Rational Expressions and Equations REQUIRED

Textbook coverage: 7.1 Simplify Rational Expressions; 7.2 Multiply and Divide Rational Expressions; 7.3 Add and Subtract Rational Expressions with a Common Denominator; 7.4 Add and Subtract Rational Expressions with Unlike Denominators; 7.5 Solve Rational Equations

---

M21-RAT-001

I can state excluded values for a rational expression.

Textbook: 7.1 Simplify Rational Expressions; 7.2 Multiply and Divide Rational Expressions; 7.3 Add and Subtract Rational Expressions with a Common Denominator; 7.4 Add and Subtract Rational Expressions with Unlike Denominators

SUPPORTING SKILLS

- I Determine excluded values from an original denominator.
  SK-RAT-RESTRICTIONS

M21-RAT-002

I can simplify a rational expression while preserving its restrictions.

Textbook: 7.1 Simplify Rational Expressions; 7.2 Multiply and Divide Rational Expressions

SUPPORTING SKILLS

- I Cancel common factors rather than individual terms.
  SK-RAT-CANCEL-FACTORS
- I Factor numerators and denominators completely.
  SK-RAT-FACTOR
- A Determine excluded values from an original denominator.
  SK-RAT-RESTRICTIONS · first introduced in Math 21

M21-RAT-003

I can multiply and divide rational expressions.

Textbook: 7.2 Multiply and Divide Rational Expressions

SUPPORTING SKILLS

- A Add, subtract, multiply, and divide fractions.
  SK-NUM-FRACTION-OPERATE · inherited from middle school
- A Cancel common factors rather than individual terms.
  SK-RAT-CANCEL-FACTORS · first introduced in Math 21
- A Factor numerators and denominators completely.
  SK-RAT-FACTOR · first introduced in Math 21
- A Determine excluded values from an original denominator.
  SK-RAT-RESTRICTIONS · first introduced in Math 21

M21-RAT-004

I can add and subtract rational expressions using a common denominator.

Textbook: 7.3 Add and Subtract Rational Expressions with a Common Denominator; 7.4 Add and Subtract Rational Expressions with Unlike Denominators

SUPPORTING SKILLS

- **A** Add, subtract, multiply, and divide fractions.
  `SK-NUM-FRACTION-OPERATE · inherited from middle school`
- **I** Combine rational expressions over a common denominator.
  `SK-RAT-COMBINE`
- **I** Determine the least common denominator of rational expressions.
  `SK-RAT-LCD`

M21-RAT-005

I can solve a rational equation and reject extraneous solutions.

Textbook: 7.5 Solve Rational Equations

SUPPORTING SKILLS

- **A** Use inverse operations to maintain an equivalent equation.
  `SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12`
- **I** Clear denominators from a rational equation.
  `SK-RAT-CLEAR-DENOMINATORS`
- **I** Check candidate solutions against denominator restrictions.
  `SK-RAT-EXTRANEOUS`
- **A** Determine excluded values from an original denominator.
  `SK-RAT-RESTRICTIONS · first introduced in Math 21`

Exponents and Radicals REQUIRED

Textbook coverage: 8.1 Simplify Expressions with Roots; 8.2 Simplify Radical Expressions; 8.3 Simplify Rational Exponents; 8.4 Add, Subtract, and Multiply Radical Expressions

---

M21-EXP-001

I can simplify expressions containing negative exponents.

Textbook: 8.3 Simplify Rational Exponents

SUPPORTING SKILLS

- I Rewrite a negative exponent using a reciprocal.
  SK-EXP-NEGATIVE
- A Apply the product rule for exponents.
  SK-EXP-PRODUCT · first introduced in Math 12
- A Apply the quotient rule for exponents.
  SK-EXP-QUOTIENT · first introduced in Math 12

M21-EXP-002

I can rewrite expressions between radical and rational-exponent forms.

Textbook: 8.3 Simplify Rational Exponents

SUPPORTING SKILLS

- I Interpret numerator and denominator roles in a rational exponent.
  SK-EXP-RATIONAL
- I Convert between radical and rational-exponent notation.
  SK-RAD-CONVERT

M21-EXP-003

I can simplify radical expressions.

Textbook: 8.1 Simplify Expressions with Roots; 8.2 Simplify Radical Expressions; 8.4 Add, Subtract, and Multiply Radical Expressions

SUPPORTING SKILLS

- A Express a whole number as a product of prime factors.
  SK-NUM-PRIME-FACTOR · inherited from middle school
- I Extract perfect-power factors from a radical.
  SK-RAD-FACTOR-PERFECT

M21-EXP-004

I can apply exponent rules to expressions with integer and rational exponents.

Textbook: 8.3 Simplify Rational Exponents; 8.4 Add, Subtract, and Multiply Radical Expressions

SUPPORTING SKILLS

- **A** Rewrite a negative exponent using a reciprocal.
  SK-EXP-NEGATIVE · first introduced in Math 21
- **D** Apply the power rule for exponents.
  SK-EXP-POWER · first introduced in Math 12
- **D** Apply the product rule for exponents.
  SK-EXP-PRODUCT · first introduced in Math 12
- **D** Apply the quotient rule for exponents.
  SK-EXP-QUOTIENT · first introduced in Math 12
- **A** Interpret numerator and denominator roles in a rational exponent.
  SK-EXP-RATIONAL · first introduced in Math 21
- **I** Apply the zero-exponent rule.
  SK-EXP-ZERO

M21-EXP-005

I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- **A** Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- **I** Require an even-root radicand to be nonnegative when determining domain.
  SK-FUNC-DOMAIN-EVEN-ROOT
- **I** Check a candidate solution for extraneous results.
  SK-RAD-CHECK
- **I** Isolate a radical expression before solving.
  SK-RAD-ISOLATE

Quadratic Functions and Equations REQUIRED

Textbook coverage: 9.1 Solve Quadratic Equations Using the Square Root Property; 9.2 Solve Quadratic Equations by Completing the Square; 9.3 Solve Quadratic Equations Using the Quadratic Formula; 9.4 Solve Equations in Quadratic Form; 9.5 Solve Applications of Quadratic Equations; 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations

---

M21-QUAD-001

I can identify key features of a quadratic function from its graph or equation.

Textbook: 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations

SUPPORTING SKILLS

- I Identify standard, factored, and vertex forms.
  SK-QUAD-FORM-IDENTIFY
- I Determine x- and y-intercepts of a quadratic function.
  SK-QUAD-INTERCEPTS
- I Determine the vertex and axis of symmetry.
  SK-QUAD-VERTEX

M21-QUAD-002

I can connect standard, factored, and vertex forms to features of a parabola.

Textbook: 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations

SUPPORTING SKILLS

- I Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
  SK-QUAD-FORM-CONVERT
- D Identify standard, factored, and vertex forms.
  SK-QUAD-FORM-IDENTIFY · first introduced in Math 21
- A Determine x- and y-intercepts of a quadratic function.
  SK-QUAD-INTERCEPTS · first introduced in Math 21
- A Determine the vertex and axis of symmetry.
  SK-QUAD-VERTEX · first introduced in Math 21
- D Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M21-QUAD-003

I can solve a quadratic equation by factoring.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- A Factor a quadratic trinomial with a nonunit leading coefficient.
  SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- A Factor a monic quadratic trinomial.
  SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- D Apply the zero-product property.
  SK-FACTOR-ZERO-PRODUCT · first introduced in Math 21
- I Determine real and complex zeros using an appropriate method.
  SK-POLY-ZEROS

M21-QUAD-004

I can solve a quadratic equation using the square-root property.

Textbook: 9.1 Solve Quadratic Equations Using the Square Root Property

SUPPORTING SKILLS

- I Apply the square-root property with both signs.
  SK-QUAD-SQUARE-ROOT
- A Extract perfect-power factors from a radical.
  SK-RAD-FACTOR-PERFECT · first introduced in Math 21

M21-QUAD-005

I can solve a quadratic equation by completing the square.

Textbook: 9.2 Solve Quadratic Equations by Completing the Square

SUPPORTING SKILLS

- A Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I Create a perfect-square trinomial by completing the square.
  SK-QUAD-COMPLETE-SQUARE

M21-QUAD-006

I can solve a quadratic equation using the quadratic formula.

Textbook: 9.3 Solve Quadratic Equations Using the Quadratic Formula

SUPPORTING SKILLS

- I Use the discriminant to predict the number and type of solutions.
  SK-QUAD-DISCRIMINANT
- I Substitute coefficients accurately into the quadratic formula.
  SK-QUAD-FORMULA
- A Extract perfect-power factors from a radical.
  SK-RAD-FACTOR-PERFECT · first introduced in Math 21

M21-QUAD-007

I can choose an efficient method for solving a quadratic equation.

Textbook: 9.1 Solve Quadratic Equations Using the Square Root Property; 9.2 Solve Quadratic Equations by Completing the Square; 9.3 Solve Quadratic Equations Using the Quadratic Formula; 9.4 Solve Equations in Quadratic Form

SUPPORTING SKILLS

- A Apply the zero-product property.
  SK-FACTOR-ZERO-PRODUCT · first introduced in Math 21
- A Create a perfect-square trinomial by completing the square.
  SK-QUAD-COMPLETE-SQUARE · first introduced in Math 21
- A Substitute coefficients accurately into the quadratic formula.
  SK-QUAD-FORMULA · first introduced in Math 21
- I Select an efficient quadratic-solving method from equation structure.
  SK-QUAD-METHOD-SELECT
- A Apply the square-root property with both signs.
  SK-QUAD-SQUARE-ROOT · first introduced in Math 21

M21-QUAD-008

I can graph a quadratic function using its vertex, intercepts, and symmetry.

Textbook: 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations

SUPPORTING SKILLS

- A Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
  SK-QUAD-FORM-CONVERT · first introduced in Math 21
- D Determine x- and y-intercepts of a quadratic function.
  SK-QUAD-INTERCEPTS · first introduced in Math 21
- D Determine the vertex and axis of symmetry.
  SK-QUAD-VERTEX · first introduced in Math 21
- D Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21

M21-QUAD-009

I can create and interpret a quadratic model for a projectile, area, or optimization problem.

Textbook: 9.5 Solve Applications of Quadratic Equations

SUPPORTING SKILLS

- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- I Interpret vertex and zeros in an application.
  SK-QUAD-MODEL-FEATURES
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12

Extension Content EXTENSION

Textbook coverage: 9.3 Solve Quadratic Equations Using the Quadratic Formula

M21-EXT-001

I can perform basic arithmetic with complex numbers.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- I Simplify integer powers of the imaginary unit.
  SK-CPLX-I-POWERS
- I Add, subtract, and multiply complex numbers.
  SK-CPLX-OPERATE

M21-EXT-002

I can express the nonreal solutions of a quadratic equation using complex numbers.

Textbook: 9.3 Solve Quadratic Equations Using the Quadratic Formula

SUPPORTING SKILLS

- A Simplify integer powers of the imaginary unit.
  SK-CPLX-I-POWERS · first introduced in Math 21
- I Express a quadratic solution with a negative discriminant in complex form.
  SK-CPLX-QUAD-SOLUTION
- A Use the discriminant to predict the number and type of solutions.
  SK-QUAD-DISCRIMINANT · first introduced in Math 21